



Analyst	Admin
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Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	2d, double precision, pressure-based, laminar, transient
Version	26.1.0-10262
Source Revision	b6f454dde8
Build Time	Feb 2 2026 16:29:27 EST
CPU	13th Gen Intel(R) Core(TM) i7-13700K
OS	Windows

Geometry and Mesh

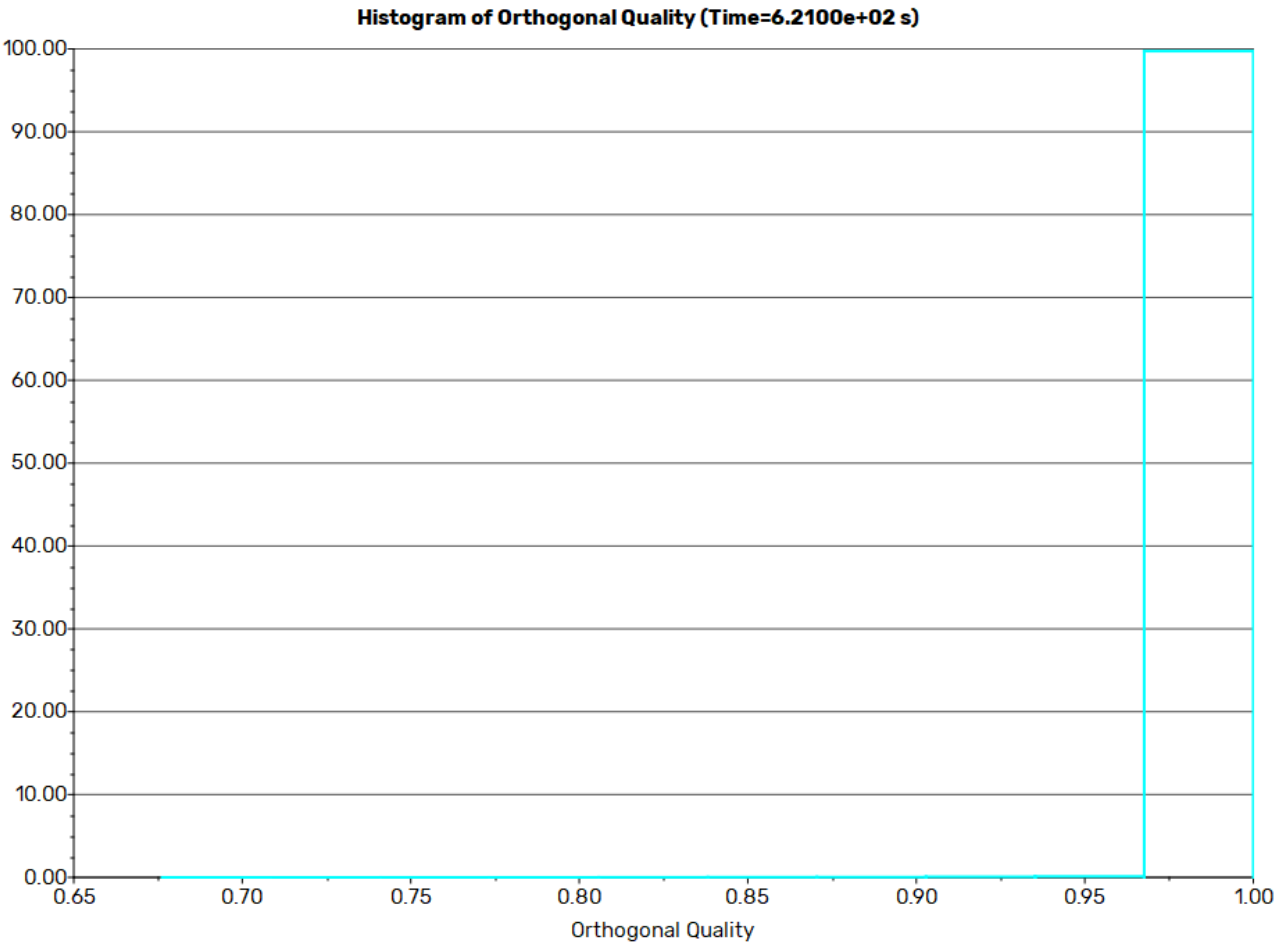
Mesh Size

Cells	Faces	Nodes
60238	120991	60753

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fff_surface	Mixed Cell	0.67627935	20.704562

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Unsteady, 1st-Order Implicit
Viscous	Laminar

Material Properties

– Fluid	
– air	
Density	1.225 kg/m ³
Viscosity	1.7894e-05 kg/(m s)
– Solid	
– aluminum	
Density	2719 kg/m ³

Cell Zone Conditions

– Fluid	
– fff_surface	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Mesh Motion?	no
Porous zone?	no

Boundary Conditions

– Inlet	
– inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute
Velocity Magnitude [m/s]	0.02
Supersonic/Initial Gauge Pressure [Pa]	0
– Outlet	
– outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
– Wall	
– side-walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
– cylinder-wall	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

Reference Values

Area	1 m ²
Density	1.225 kg/m ³
Depth	1 m
Enthalpy	0 J/kg
Length	1 m

Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fff_surface

Solver Settings

– Equations	
Flow	True
– Numerics	
Absolute Velocity Formulation	True
– Unsteady Calculation Parameters	
Number of Time Steps	4500
Time Step Size [s]	0.1
Max Iterations/Time Step	20
– Under-Relaxation Factors	
Pressure	0.3
Density	1
Body Forces	1
Momentum	0.7
– Pressure-Velocity Coupling	
Type	SIMPLE
– Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1

Maximum Static Temperature [K]	5000
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Run Information

Number of Machines	1
Number of Cores	10
Case Read	9.42 seconds
Iteration	290.912 seconds
AMG	114.944 seconds
Virtual Current Memory	2.56683 GB
Virtual Peak Memory	2.58664 GB
Memory Per M Cell	31.5057

Solution Status

Flow Time: 621

Time Step: 6210

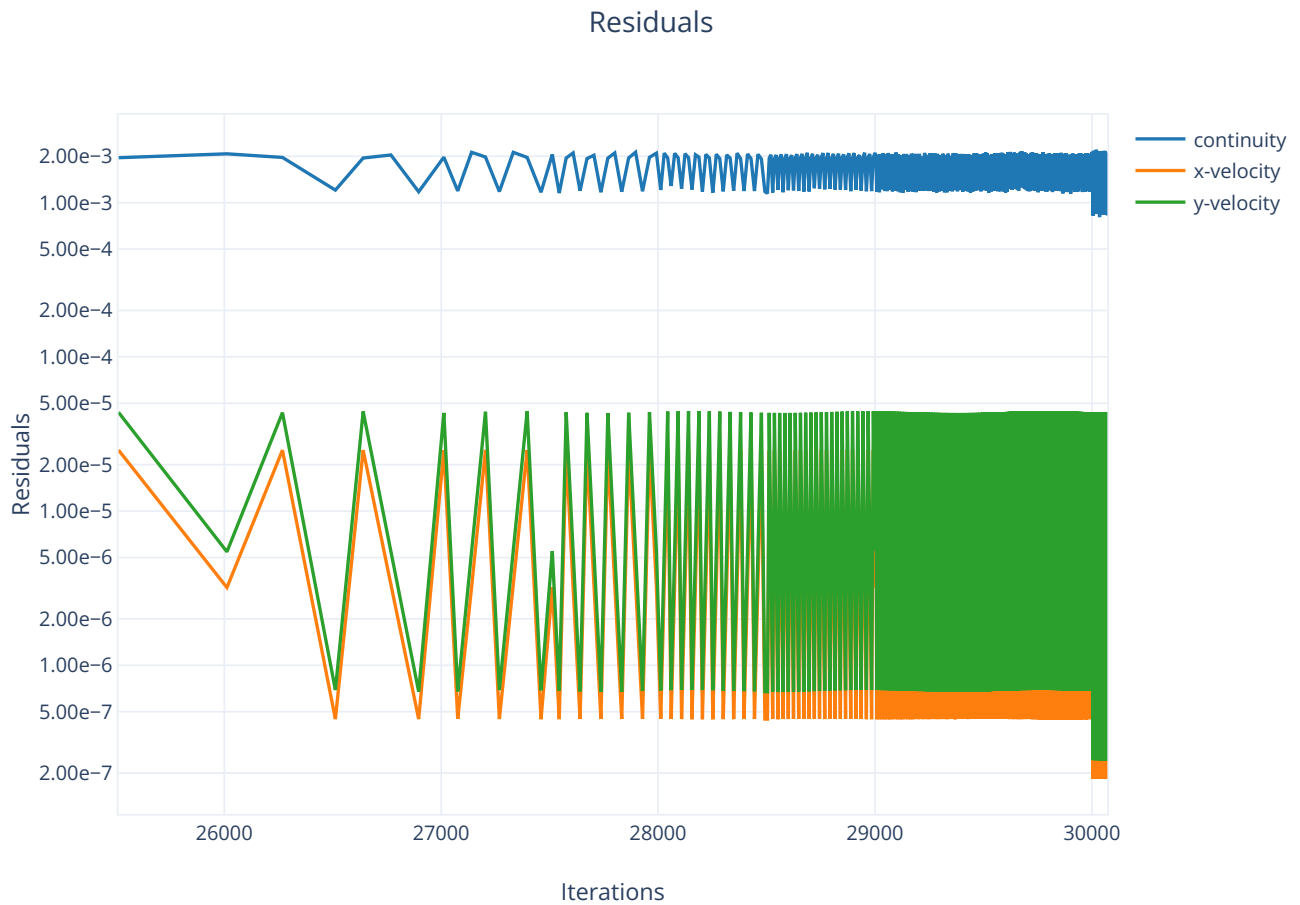
	Value	Absolute Criteria	Convergence Status
continuity	0.0008251056	0.001	Converged
x-velocity	1.835341e-07	0.001	Converged
y-velocity	2.413879e-07	0.001	Converged

Report Definitions

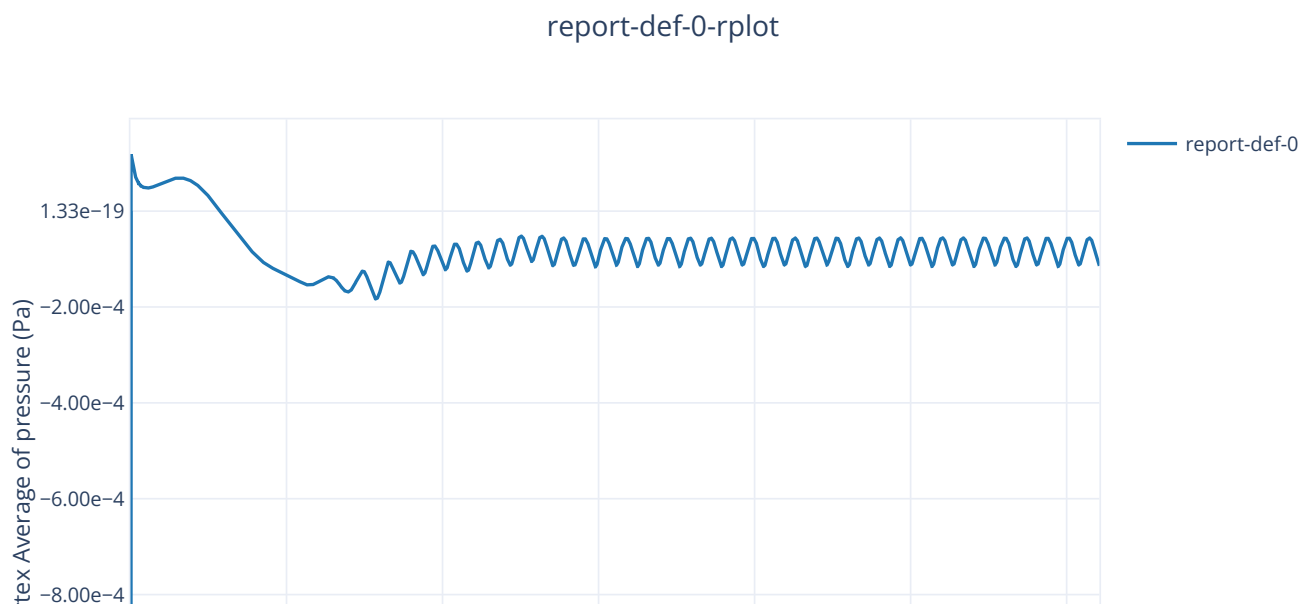
report-def-0	-0.0001142246	Pa
delta-time	0.1	s
iters-per-timestep	6	
flow-time	621	s

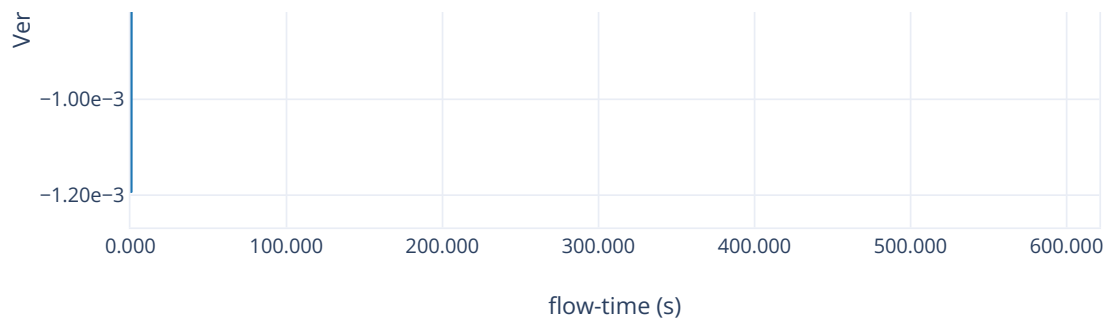
Plots

Residuals



report-def-0-rplot

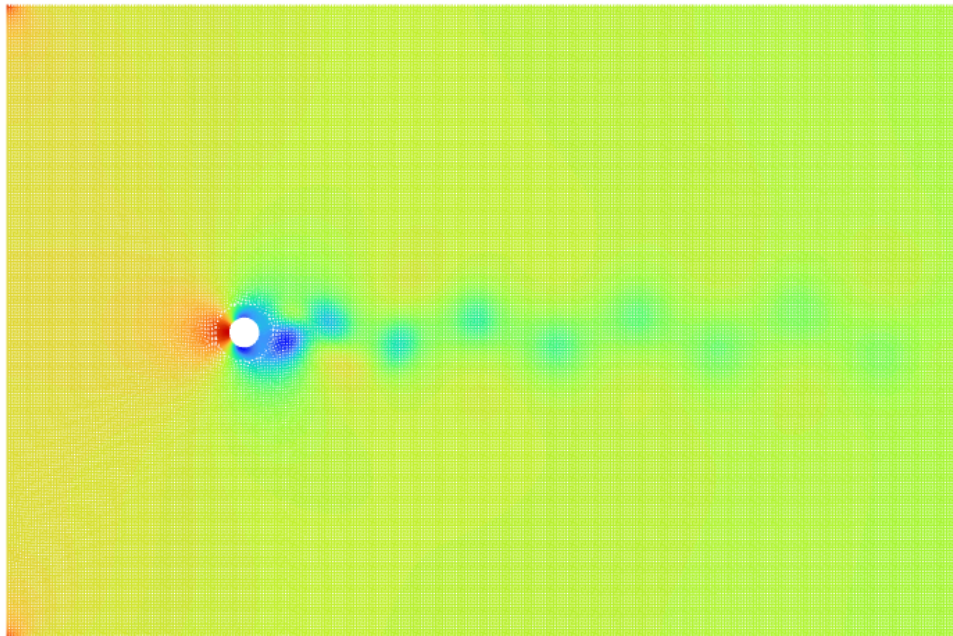
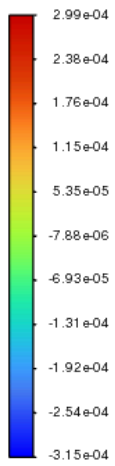




Contours

contour-1

Static Pressure
[Pa]



contour-1